EduSimplify - Demo Study Material

Topic: Photosynthesis & Cell Biology

Introduction

Photosynthesis is the biochemical process by which autotrophic organisms, primarily plants, algae, and certain prokaryotes, convert light energy into chemical energy stored in glucose molecules. This process occurs in the chloroplasts of plant cells, specifically utilizing the pigment chlorophyll to absorb photons from the visible spectrum of electromagnetic radiation.

The overall chemical equation for photosynthesis is:
$6CO_2 + 6H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2$

This reaction can be subdivided into two primary stages:
1. The Light-Dependent Reactions (also known as the Hill Reaction)
2. The Light-Independent Reactions (the Calvin-Benson Cycle)

Section 1: Light-Dependent Reactions

The light-dependent reactions occur in the thylakoid membranes of the chloroplast. When photons strike the chlorophyll molecules embedded in Photosystem II (PSII), electrons are excited to a higher energy state and are subsequently transferred through the electron transport chain (ETC).

Key processes in this stage:
- Water Photolysis: Water molecules are split (photolysis) releasing O2 as a byproduct, electrons, and hydrogen ions (protons).
- Electron Transport Chain: Excited electrons pass through a series of protein complexes (plastoquinone, cytochrome b6f, plastocyanin), releasing energy used to pump protons across the thylakoid membrane.
- ATP Synthesis: The proton gradient drives ATP synthase (chemiosmosis) to produce ATP from ADP and inorganic phosphate.
- NADPH Formation: At Photosystem I (PSI), electrons reduce NADP+ to NADPH, a critical electron carrier for the Calvin Cycle.

The products of the light-dependent reactions are:
ATP, NADPH, and O2 (released into the atmosphere).

Section 2: The Calvin Cycle (Light-Independent Reactions)

The Calvin Cycle occurs in the stroma of the chloroplast and does not directly require light, but depends on the ATP and NADPH produced in the light reactions.

The cycle has three main phases:

Phase 1 - Carbon Fixation:
The enzyme RuBisCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase) catalyzes the attachment of atmospheric CO_2 to a 5-carbon molecule called RuBP (ribulose bisphosphate), producing an unstable 6-carbon compound that immediately splits into two 3-carbon molecules of 3-PGA (3-phosphoglycerate).

Phase 2 - Reduction:
ATP and NADPH are consumed to convert 3-PGA into G3P (glyceraldehyde-3-phosphate), a 3-carbon sugar. G3P is the primary carbohydrate product of photosynthesis and serves as a building block for glucose and other organic molecules.

Phase 3 - Regeneration of RuBP:
Most G3P molecules are used to regenerate RuBP using ATP, allowing the cycle to continue. For every 3 CO_2 molecules fixed, one net G3P is produced.

Significance:
Photosynthesis is the foundation of nearly all food chains on Earth, converting solar energy into chemical energy that sustains life. It also regulates atmospheric CO_2 levels, playing a vital role in Earth's carbon cycle.